The Twist, version 5.2: errata and post-publication results

Danny Lee Eldridge (independent researcher) · 14 November 2026 · errata to “The Twist: a no-scale supergravity field for inflation and a thawing dark-energy tail” (v5, DOI 10.5281/zenodo.24210532). Supersedes the v5.1 errata of the same day.

This note lists corrections and new results found after v5 was published on 13 November 2026. The v5 paper and exploratory note are unchanged; read them together with this note. Notes labelled D… are in theory/pending_docs/D72_D75_weak_spots.md of the archived repository (branch unify-bg); locked rules are in field_tests/rules/. AI-assisted.

Summary of what changed

1. **The fifth-force shield fails at one loop for the thawing tail (item 16).** A rolling field with $K \supset -3 \ln(X + \bar{X})$ moves the Standard Model gauge couplings by $d(1/\alpha_a)/d \ln x = -b_a/4\pi$ (Kaplunovsky–Louis; fixed by rescaling anomalies), so $d \ln \alpha_{em}/d \ln x \approx 7\times10^{-3}$. Seven escape routes were tested, two with locked inflation tests.
2. **Adopted reading: TWIST-I (item 18).** The Twist field rolls as in v5 through inflation, reheating and matter genesis. It is only $x \approx 3$ at matter freeze-out and converges to $x \approx 29.8$ by $T \sim 10^{11}$ GeV. It is then captured at a point $x_E \approx 27$–29.5 by $W \supset (T - T_E)(\lambda M_P \chi\bar{\chi} + k Y)$: χ traps the field, and a weight-3 stabiliser Y gives an exact, untilted $V = k^2|T - T_E|^2$ with its minimum at the trapping point. The capture costs almost no entropy ($\Delta \approx 1 + O(10^{-3})$), so no refits are needed. $\Lambda = V_{Twist}(x_E) \propto e^{-2n}$ keeps “Λ from the cliff” ($n \approx 124.1$, no effect on n_s or r). Dark energy is exactly $w = -1$. The original thawing tail is kept only as a conditional variant. Locked rule: PREREG_TWISTI (+ deviations D1–D9); write-up: papers/twist_v5/TWISTI_CAPTURE.md. The UV origin of the capture sector is not derived.
3. **TWIST-I’s CMB predictions are Starobinsky-class (item 19).** n_s 0.9642–0.9654 and r 0.0035–0.0038 at $N^* = 55.7$–57.6. The built-in kination raises N^* toward, but not into, the $N^* \gtrsim 58$ that ACT compatibility needs. The distinctive content is N^*, the matter mechanism and its isocurvature, and the 100–200 TeV dark matter.
4. **The Twist angle is heavy after capture (item 20).** This removes the axion-overproduction clause and the eV-branch angle relic. Strong CP still needs a separate axion.
5. **Locked tests since v5 (all pass):** the matter yield is 1.25× observed. The isocurvature sensitivity is 3.70, and 3.68 under the alternative coupling reading. At the alternative fit point $m_\sigma = 3\times10^{10}$ GeV it is 1.2, giving a margin of 4.2×.
6. **Updated costs and calendar:** the full-fit H_0 for the tail is 66.8 (ΛCDM 68.0; back to ΛCDM under TWIST-I). The n_s tension depends on the dark-energy background. CMB-S4 is cancelled; the r test rests on the expanded Simons Observatory ($\sigma_r \approx 1.2\times10^{-3}$) and LiteBIRD ($\delta r = 0.001$, launch JFY 2032). DESI and Euclid dark-energy results are due in 2027.
7. **Still running at the time of this note:** a free-neutrino-mass fit (locked, PREREG_D355_MNU_FREE) and the kinetic-share

consistency run (D367; a preliminary reading is in D367a). Their results go to the repository.

Itemised corrections (as queued for v6)

1. Section 5.2 (Gravitino) and assumption 4: replace the text (D267, D269)

The current v5 text says: “Supersymmetry breaking comes from a separate, unspecified sector … m_3/2 = 10^{11}–10^{14} GeV”.

Replace it with:

> W = √3 m S T vanishes at S = 0, so the inflaton sector does not fix m_3/2. In this field space, any supersymmetry-breaking superpotential gives m_3/2 = |W|/(2x)^{3/2} on real T [Zwirner, arXiv:2504.06190; Ellis et al., arXiv:1809.10114].
>
> **A constant W_0 is excluded.** It changes m_3/2 by a factor 5.7×10^{8} between the CMB point and today. Either the gravitino is super-Planckian during inflation, or it is $\lesssim 3\times10^{4}$ GeV today. In the second case, thermal production at T_RH ≈ 5.6×10^{14} GeV violates nucleosynthesis bounds.
>
> Two options remain:
>
> **(i) A light gravitino from low-scale gauge mediation.** m_3/2 ≈ 0.8–4.7 eV today, between the Higgs-mass lower limit [Ibe & Yanagida, arXiv:1608.01610] and the cosmological upper limit [Osato et al., arXiv:1601.07386]. Along the Twist’s history it is 0.5–3 GeV during inflation ($\ll$ H) and a few keV at the cliff, so it is harmless. Keeping V ≈ 0 at every x makes the SUSY-breaking scale slide as x^{−3/2} too. Superpartner masses would then change by ~30% between nucleosynthesis and today, shifting α_em by $\sim10^{-3}$ through thresholds (estimate). So this option needs the shield to cover the SUSY-breaking scale seen by the Standard Model (D272).
>
> **(ii) W ⊃ w T^{3/2} plus a constant uplift.** This gives an x-independent m_3/2. - It leaves the plateau intact: a full supergravity check finds V changes by ≤ 0.6% for m_3/2 ≤ 10^{13} GeV. - It gives the angle a tachyonic mass $-2\ m_{3/2}^2$. Kähler-only stabilisers do not cure this: m_θ^2 stays $-2\ m_{3/2}^2$ for any $b(T-\bar{T})^4$ or $b(T-\bar{T})^2$ term (D270). A cure needs the uplift sector to couple to the angle, which makes the angle heavy (~m_3/2). - It needs m_3/2 $\lesssim 6\times10^{8}$ GeV, so that the angle stays light while matter forms. - It needs m_3/2 $\gtrsim 10^{8}$ GeV, so that gravitino decay releases $\lesssim$ 15% of the energy. - It therefore works only for m_3/2 ≈ 10^{8}–10^{9} GeV (estimate), with R-parity violation.
>
> Neither option is derived. Adopt (i).

2. Section 2.2 (coupling): result of the alternative reading

Add to §2.2: “Coupling instead to ∂(y/x) (the 1/γ_BI = y/x reading) lowers the matter yield by a factor 0.44 (D268/D297). The fitted hidden mass then moves from m_σ ≈ 7.5×10^{10} to ≈ 3.0×10^{10} GeV. The

mechanism works under either reading."

3. Table of assumptions, item 4

Replace the item with: "Supersymmetry breaking: a near-massless gravitino with superpartner masses not tied to m_3/2 (adopted); W ∝ T^{3/2} plus uplift needs the uplift sector to stabilise the angle and works at best for m_3/2 ≈ 10^8–10^9 GeV; a constant W_0 is excluded."

4. Section 5.1 (fifth force): add precedents (D273)

Add after the shield paragraph:

> In generic supergravity quintessence, every MSSM soft term and fermion mass depends on the field through e^{K/2}, and Cassini then requires a tuned cancellation [Brax & Martin, hep-th/0605228]. The shield must therefore also keep the Standard Model's supersymmetry-breaking scale independent of x. The no-scale isometry protects the field's own lightness from gravitino-induced masses [Brax, Davis & Sakstein, arXiv:1212.4392], but not its couplings to matter.

5. Exploratory note §1.3–1.4: locked PC results (D277)

- **m_σ.** Replace "confirmation pending" with: "the locked confirmation run gives 1.25× observed at m_σ = 5.8×10^{10} GeV, the same in TWIST-K; entropy dilution ×4.6".
- **§1.4 Isocurvature.** Replace "unverified" with: "the locked sensitivity run gives |d ln Y/d ln σ_0| = 3.70, which passes the locked limit of 5. S_B ≈ 3.7×10^{-5} against a bound of 5×10^{-5} at f = 0.1 M_P is marginal, and requires f ≳ 0.1 M_P. The yield is very sensitive to the angle's starting value (y_0 = 0.01 multiplies it by about 9000), so the heavy angle during inflation is essential." The earlier "8× below the bound" is withdrawn.

6. Gravitino consequences (D279)

Add to §5.2 option (i): "The gravitino is a thermal warm relic, making up m_3/2/86 eV ≈ 1–5% of the dark matter, with ΔN_eff ≈ 0.02–0.05. The main dark-matter candidate is unaffected."

7. A new stated assumption: the reheating field χ has modular weight 3/2 (D286)

Add to the assumptions list:

> **The reheating field χ enters the Kähler potential as $|\chi|^2/(T + \bar{T})^{3/2}$.** Its physical mass is then independent of x, and its one-loop correction to the plateau is negligible, as D156 assumes.

> The other placements fail: - Inside the no-scale log, the one-loop term would be an order-one $\propto 1/x$ deformation ($dV/V \approx 7$ at the CMB point). - With a flat Kähler term, it would be larger still.

8. Option (i) needs a non-minimal SUSY-breaking sector (D291)

Add to §5.2:

> The 0.8–4.7 eV window is open only in non-minimal models, such as vector-mediated D-term breaking [Hook, McGehee & Murayama, arXiv:1801.10160] or strongly coupled messengers [Lu & Shirai, arXiv:1712.02359]. Minimal gauge mediation is excluded. A cosmic-shear bound below ~1 eV would remove this option.

9. A supersymmetry-breaking sector with x-independent scales (D292; supersedes the two-option text in item 1)

Replace the §5.2 options with:

> **$W \supset w\, T^{3/2} + \mu^2 X$, with the Polonyi field X at modular weight 3:** $K \supset |X|^2 (T + \bar{T})^{-3} [1 + c((T - \bar{T})/(T + \bar{T}))^2]$.
>
> - Along real T, exactly: $V = \mu^4 - 3w^2/8$, the same for every x. A single cosmological-constant tuning sets it to zero.
> - $m_{3/2} = w/2^{3/2}$ and $F_X = \mu^2$, both x-independent.
> - The angle mass is $(4c - 2)\, m_{3/2}^2$, so the angle is stable for $c > 1/2$.
> - With messengers at modular weight 3/2, the gauge-mediated soft masses do not slide. This removes the α_{em} drift of D289.
> - The eV gravitino window (D291) is realised in this sector.
> - The modular weights are assumed, not derived.

10. Condition on the SUSY-breaking sector (D294)

Add to §5.2:

> $X = 0$ is shifted by a Polonyi tadpole to $X \approx 2\sqrt{6}/(12q - 2)\, M_P$. That shift is x-independent in canonical units. The sector must also give the sgoldstino a mass far above $m_{3/2}$ ($q \gtrsim 10^7$); otherwise its late oscillations overclose the universe. This holds for dynamical SUSY breaking at $\sqrt{F} \sim 10^5$ GeV, but not for a bare Polonyi field.

11. Angle relic in the eV-gravitino branch (D298–D299)

Add to the exploratory note:

> With the late angle mass from §5.2 (~1–5 eV), the angle's leftover displacement after matter formation ($\varphi_i \approx 1.1\times10^{11}$ GeV, computed) gives a cold, ALP-like relic. It makes up roughly 12–28% of the dark matter (order-of-magnitude estimate), so it does not overclose. This is a consequence that could be tested; it is not yet locked.

12. Literature framing for the string-home limitation (D302, D304)

Add to §7:

> No controlled string regime is known for a thawing tail like this one. Cicoli et al. (Fortsch. Phys. 2022) find no slow-roll region in the parametrically controlled tree-level type II or heterotic moduli space; heterotic runaways give $\varepsilon = 4$. For minimal Z_3/Z_7 orbifolds, the modulus-independent gauge couplings needed by the shield are excluded by duality-anomaly matching [Ibáñez & Lüst, hep-th/9202046], so they require extra charged matter.

13. Why the tail field cannot be a heterotic volume modulus (D309)

Add to §7:

> In the weakly coupled heterotic string, $\alpha_{GUT} = e^{2\varphi}\,\alpha'^3/(16\pi V)$ [Witten, hep-th/9602070]. Keeping the gauge couplings fixed while the volume grows (as the shield requires) therefore drives the string coupling strong. The Twist's x cannot be a heterotic volume modulus. The strong-coupling continuation, heterotic M-theory, moves the Standard Model coupling along its matter-neutral direction (D208).

14. Exploratory note: a candidate string home (papers/twist_v5/G2_HOME_PROPOSAL.md, D309–D315)

Add a short section summarising the G_2 block-split proposal, and list question 12 among the open questions for string theorists.

15. n_s inconsistency is background-conditional (D223, D329)

Add to the n_s discussion:

> The NS-SPA inconsistency arises from CMB + BAO jointly. For Planck + DESI, freeing (w_0, w_a) returns $n_s = 0.9638 \pm 0.0038$, the CMB-only value [arXiv:2512.05108]. The Twist's own, milder dark-energy evolution does not do this: our NS-SPA fit with the Twist background still prefers $n_s \approx 0.973$–0.976. The verdict therefore depends on how the BAO–CMB tension resolves (DESI DR3, Simons Observatory).

16. The shield as a one-loop no-go (D323–D334)

Replace the shield paragraph's "assumption" wording with a precise statement:

> If the dark-energy field X enters as $K \supset -3 \ln(X + \bar{X})$ (which makes the plateau and the tail), the holomorphic f and Yukawas are X-independent, and the matter metrics are universal, then at one loop the Standard Model's Yukawas and gauge couplings cannot both be X-independent. Keeping Yukawas fixed gives $d(1/\alpha_a)/d \ln x = -b_a/4\pi$ [Kaplunovsky–Louis; local-model form in arXiv:0901.4350; general hidden-sector statement in arXiv:1811.10633], so $d \ln \alpha_{em}/d \ln x \approx 7\times10^{-3}$. Non-universal weights with a Giudice–Masiero μ need a Green–Schwarz term. The dilaton version of that shifts the no-scale α by ∓ 0.014 and spoils inflation (locked test). A twisted-moduli version passes inflation only with group-dependent couplings $\propto b_a$, which no known model provides. The shield is therefore an assumption beyond known string constructions, with a precisely stated requirement.

17. CMB-S4 cancelled (D346)

DOE and NSF ended support for CMB-S4 in July 2025. In the paper's tests table, replace "Simons Observatory, LiteBIRD, CMB-S4 class" with "Simons Observatory (and upgrades), BICEP Array, LiteBIRD". In the exploratory note §5, replace "LiteBIRD and CMB-S4" with "LiteBIRD and Simons Observatory". Add: with $r \approx 0.0036$, the expanded Simons Observatory (six SATs to 2035, $\sigma_r \approx 1.2\times10^{-3}$ baseline, arXiv:2512.15833) reaches ~3σ, and LiteBIRD ($\delta r = 0.001$, launch JFY 2032, arXiv:2406.02724) is comparable; an r decision is expected ~2032–35.

18. The split model TWIST-I (Danny's choice, 14 Nov 07:08; PREREG_TWISTI + D1–D9; full write-up papers/twist_v5/TWISTI_CAPTURE.md)

Present v6 with two readings.

a. **TWIST-I (adopted).** The Twist field rolls exactly as in v5 through inflation, reheating and matter genesis. It is only $x \approx 3$ at matter freeze-out, and converges to $x \approx 29.8$ by $T \sim 10^{11}$ GeV (D367a). It is captured at a point $x_E \approx 27$–29.5 that it crosses at $T \approx 10^{11}$–2×10^{12} GeV, after the matter yield has settled. The construction is $W \supset (T - T_E)(\lambda M_P \chi\bar{\chi} + k Y)$, with χ at modular weight 3/2 and Y at weight 3:

- χ is massless at T_E and traps x there (moduli trapping, hep-th/0403001).
- Y gives $V = k^2|T - T_E|^2$ exactly (no e^K tilt), so the minimum coincides with the trapping point by construction (D373–D375).
- x and the angle get masses $\gtrsim 10^5$ GeV and decay before BBN.
- Loop tilts are negligible (D374).
- Bounds: $0.004 \lesssim \lambda \lesssim 0.03$ (D377).
- The capture energy is tiny ($\Delta \approx 1 + O(10^{-3})$, D384), so no refits are needed.

The vacuum energy after capture is V_Twist(x_E) ∝ e^{−2n}, so “Λ from the cliff” is kept, with n ≈ 124.1 instead of 124.6 and no effect on n_s or r (D383). Dark energy is exactly w = −1 (no tail). With the angle heavy, the axion-overproduction clause and the eV-branch angle relic disappear (D382; item 20). This removes the shield problem (item 16) and the H_0 cost. It is not derived from a UV theory: the factor (T − T_E) and the weights are assumed.

b. **The original tail** is kept as a conditional variant, valid only if the shield no-go (item 16) is evaded.

The tests table splits accordingly: r, n_s and β apply to both; w = −1 applies to (a); the tail tests (WATCH-PURE-DR3, WA-CEILING, GROWTH, E_G) apply only to (b).

19. TWIST-I’s CMB predictions are Starobinsky-class (D368)

Add to the TWIST-I description: “At N* = 55.7–57.6, the Twist’s n_s and r agree with the α = 1 attractor formulas ($n_s = 1 - 2/N$, $r = 12/N^2$) to within 10^{-4} and 3×10^{-4}. The inflationary test therefore probes the plateau class and, through N*, the Twist’s reheating history (its built-in kination phase raises N* to 55.7–57.6, toward but not into the $N^* \gtrsim 58$ that refined Starobinsky analyses need for ACT compatibility, arXiv:2504.20757; D370). The model-specific content is in N*, the matter mechanism and its isocurvature, and the dark-matter window.”

20. Under TWIST-I the Twist angle is heavy (D382)

The weight-3 stabiliser gives $V = k^2[(x - x_E)^2 + y^2]$, so the angle y gets the same mass as x ($\gtrsim 10^5$ GeV) and decays before BBN. In the exploratory note §7, replace the strong-CP line’s “the angle would overproduce axion dark matter by $\sim4\times10^5$” with “the angle is heavy after capture and is not the QCD axion; a separate post-inflation axion (40–600 μeV) is still needed”. The eV-branch angle relic (item 11) does not arise under TWIST-I.